

Alejandro Ruiz-Martinez

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EDUCATION

Imperial College London | *MEng Computer Science* 2013-2017

- First-class honors (GPA 3.9/4.0); awarded Imperial Prize for Excellence in final year project on distributed consensus
- Led Systems Architecture club (60+ members); organized weekly seminars with engineers from Citadel and Google on low-latency systems design

University of Cambridge | *Postgraduate Diploma in Advanced Computing* 2017-2018

- Distinction (92/100) in Distributed Systems and Advanced Algorithms; thesis published in ACM SIGOPS Workshop on Research on Cloud Computing
- Teaching assistant for 3 courses on concurrency and distributed systems; mentored 25 undergraduates in systems design projects

EXPERIENCE

Stripe | *Senior Backend Engineer, Payments Infrastructure* 2021-Present

- Architected new payment processing pipeline reducing latency from 450ms to 89ms p99 (80% improvement) for 8M+ daily transactions; directly enabled 3 new partnerships worth \$150M ARR
- Owned ledger microservice handling \$2.3T in annual volume; optimized database queries reducing CPU utilization by 42%, saving \$680K/year in infrastructure costs across 12 regions
- Led cross-functional team of 6 engineers (backend, payments, security) on idempotency framework; prevented \$4.2M in duplicate transaction losses, reduced fraud losses by 23% YoY

Wise (formerly TransferWise) | *Backend Engineer, Core Payments* 2019-2021

- Redesigned multi-currency settlement engine to support real-time processing; increased throughput from 45K to 320K transactions/hour, handling 8 currency corridors across 190+ countries
- Implemented distributed tracing infrastructure using Jaeger/Open telemetry across 120+ microservices; reduced mean time to resolution (MTTR) for production incidents from 47min to 12min
- Built redundancy layer for foreign exchange rate feeds; achieved 99.99% availability (4 minutes downtime/month) and reduced FX quote errors by 87%, protecting \$120M daily forex volume

Cloudflare | *Backend Engineer, Billing & Payments* 2018-2019

- Developed webhook delivery system processing 600M events/month with 99.95% guaranteed delivery; implemented exponential backoff with intelligent retry logic reducing customer complaint tickets by 64%
- Optimized billing calculation engine handling 2.1M active accounts; reduced monthly batch job runtime from 8.5 hours to 1.2 hours through algorithmic improvements and parallelization
- Migrated legacy payment processor integration (Stripe) for 450K+ customers with zero downtime; coordinated testing of 2.3M transactions, achieved 0 data loss

Jane Street | *Backend Engineer, Clearing & Settlement* 2017-2018

- Built real-time position reconciliation system matching 15B+ trades daily with 100ms latency; detected booking discrepancies affecting \$2.1M and prevented settlement fails worth \$8.4M
- Optimized order routing algorithm improving execution quality by 1.3 basis points on average; generated incremental trading P&L of \$12.8M annually
- Designed circuit breaker and failover mechanisms for NYSE/NASDAQ connectivity; achieved 99.998% uptime during critical trading windows across 50+ trading pairs

PROJECTS

PaymentVault | *Go, PostgreSQL, gRPC, Protocol Buffers, Redis* 2022-Present

- Open-source encrypted payment token storage service with compliance features (PCI-DSS, SOC2); 1,200+ GitHub stars, 28K monthly downloads, used by 12 fintech companies processing \$500M+ annually
- Achieved sub-millisecond token lookup (p99: 0.8ms) through optimized in-memory caching and database indexing; benchmarks show 450K ops/sec throughput on standard hardware

TxnReplay | *Python, Kubernetes, PostgreSQL, Prometheus, Grafana* 2021-2023

- Transaction replay and debugging tool for payment systems; 890 GitHub stars, used internally at 3 major fintech firms for forensic analysis; reduced payment dispute investigation time by 71%
- Processed 50M+ transaction records in replay tests; memory-efficient streaming design handles 10GB datasets with <512MB peak memory, supporting 99.2% test coverage for legacy payment flows

DistributedLedger | *Rust, DynamoDB, EventSourcing, Tokio async runtime* 2020-2022

- CRDT-based distributed ledger for microservices; 650+ GitHub stars, 15K monthly downloads from package managers; consistent replication across 7 data centers with eventual consistency guarantees
- Benchmarks demonstrate 89% reduction in reconciliation time (from 4.2 hours to 28 minutes) vs. traditional approaches; write latency: 18ms p99, 240K writes/sec sustainable throughput

EXTRACURRICULAR

QCon London & GOTO Conferences | *Conference Speaker (3 talks)* 2021-2024

- Spoke on 'Building Resilient Payment Systems at Scale' (2023, 800+ attendees); talk published on InfoQ with 52K views, invited to present at 4 additional conferences including QCon San Francisco
- Co-authored workshop 'Distributed Transactions in Payments Infrastructure' reaching 200+ engineers; workshop rated 4.8/5.0 stars (180 responses); mentored 12 participants who later implemented learnings in production

Open Payments Community Foundation | *Core Maintainer & Technical Advisor* 2019-Present

- Lead maintainer for OpenPayments specification; 450+ GitHub stars, 80+ contributors across 15 countries; specification adopted by 22 fintech companies including 2 unicorns
- Mentored 18 junior engineers on systems design and payments architecture; 6 mentees promoted to senior roles; organized monthly technical workshops averaging 45+ attendees

TECHNICAL SKILLS

Languages – Go, Python, Rust, SQL, Java, JavaScript/TypeScript

Technologies – PostgreSQL, Redis, DynamoDB, Kafka, gRPC, Protocol Buffers, Distributed Systems, Event Sourcing, CRDT, Microservices Architecture, Real-time Processing, Message Queues, Idempotency Patterns

Tools – Kubernetes, Docker, Prometheus, Grafana, DataDog, Jaeger, Git, GitHub, Jira, Terraform, AWS (EC2, RDS, DynamoDB, S3, Lambda), GCP